Public-key cryptography

Cyber security · University · Lecture 2



In this lecture

- Run Diffie–Hellman.
- Explain RSA encryption and signatures.
- Expose the man in the middle.

Remember

- Easy one way, hard back.
- Encryption ≠ authentication.
- Sign the exchange.

Alice, Bob and Eve

1. Warm-up (5 min): How is it possible?
2. Mix (10 min): Paint analogy.
3. Exchange (20 min): On paper.
4. Attack (20 min): Mallory joins.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Why couldn’t Eve find the key?
2. What did signatures add?
3. Which misconception did you hear today?

Exit ticket

- Say what encrypting to Bob’s public key does and does not guarantee.